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Antioxidant activities of peel, pulp and seed fractions of common fruits as determined by FRAP assay

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Abstract

The antioxidant activities of peel, pulp and seed fractions of 28 fruits commonly consumed in China were determined using the ferric reducing/antioxidant power assay (FRAP assay). The contribution of vitamin C to the antioxidant activity of fruit pulps was also calculated. The results showed that hawthorn pulp had the highest FRAP value among all fruit pulps and followed by date, guava, kiwifruit, purple mulberry, strawberry, white pomegranate, lukan and honey tangerine pulps and etc. Most of fruit peel and seed fractions were stronger than the pulp fractions in antioxidant activity based on their FRAP values. The contribution of vitamin C to the FRAP value of fruit pulps varied greatly from fruit to fruit as calculated. We concluded that peel and seed fractions of some fruits, such as pomegranate peel, grape seed, hawthorn peel, longan and lychee seeds possessed relatively high antioxidant activity and might be rich sources of natural antioxidants. © 2003 Elsevier Inc. All rights reserved.

Keywords: Antioxidant activity; Fruit peel; Pulp and seed

1. Introduction

Reactive oxygen species (ROS) and reactive nitrogen species (RNS) are constantly generated in vivo for physiological purposes and often over-produced in pathological conditions, resulting in oxidative stress. To protect their possible damages to biological molecules, especially to DNA, lipids and proteins, all oxygen-consuming organisms are

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endowed with a well-integrated antioxidant system, including enzymatic and non-enzymatic components. The superoxide dismutase (SOD), glutathione peroxidase (GSHPx) and catalase are the major antioxidant enzymes frequently mentioned in the literature. The non-enzymatic components consist of macromolecules, such as albumin, ceruloplasmin and ferritin as well as an array of small molecules, such as vitamin C, E, β -carotene, reduced glutathione [1,2]. The fruits are rich sources of various vitamins, minerals and fibers required by human body for optimal health. In the recent years, more attention has been paid to the antioxidants contained in fruits because epidemiological studies revealed that high fruit intake was associated with reduced mortality and morbidity of cardiovascular disease and some types of cancer and one of possible mechanisms was attributed to the antioxidant activity presented by the fruits [3,4]. Besides classical antioxidants including vitamin C, E and β -carotene, phenolic compounds had been identified as important antioxidants contained in fruits. Some phenolic compounds are even more powerful as antioxidants than vitamin C, E in vitro and significantly bioavailable as demonstrated by animal and human studies [5,6,7,8,9,10]. However, fruits are diverse in antioxidant composition and antioxidant activity and those with high antioxidant activity generally contain more antioxidants [11]. Interestingly, the peel and seed fractions of some fruits possess higher antioxidant activity than the pulp fractions. For example, grape seeds had been demonstrated to be higher than grape pulps in antioxidant capacity and were rich sources of proanthocyanidins, which were very effective in scavenging various ROS [12]. Another example is the pomegranate peel. Its methanolic extracts exhibited higher antioxidant activity in various in vitro models compared to seed extracts [13]. Therefore, the peel and seed fractions of fruits may potentially contain more antioxidants quantitatively or qualitatively than the pulp fractions.

In the present study, the ferric reducing/antioxidant power assay (FRAP assay) was employed [14] and the FRAP value of peel, pulp and seed fractions of fruits commonly consumed in China was determined in an attempt to make a systematic comparison among their antioxidant activities and identify the fractions with high antioxidant activity for further studies.

2. Materials and methods

2.1. Sample preparation

A total of 28 fruits were purchased from local markets. There were hawthorn, date, guava, kiwifruit, purple mulberry, strawberry, white pomegranate, “Lukan” tangerine, “Honey” tangerine, orange, lemon, cherry, longan, “Banana flavored” apple, pineapple, banana, plum, lychee, kumquat, “Red rose” grape, pomelo, mango, “Jiubao” peach, apricot, “Hami” melon, “Duck” pear, “Jingxin #1” watermelon and persimmon. All fruits were first flushed by tap water and then washed in distilled water for three times before the peel, pulp and seed fractions were carefully separated. A portion of 1~5 gram was weighed and ground in a mortar after addition of distilled water (1:9 w/v). The homogenate was centrifuged at 6000g for 10 min. The supernatant was recovered and used directly for FRAP assay without storage.

2.2. FRAP assay

The procedure described by Benzie and Strain was followed [14]. The principle of this method is based on the reduction of a ferric-tripyridyltriazine complex to its ferrous, colored form in the presence of antioxidants. Briefly, the FRAP reagent contained 2.5 ml of a 10 mmol/L TPTZ (2,4,6- tripyridyl-s-triazine, Sigma) solution in 40 mmol/L HCl plus 2.5 ml of 20 mmol/L FeCl_3 and 25 ml of 0.3 mol/L acetate buffer, pH 3.6 and was prepared freshly and warmed at 37°C. Aliquots of 40 μl sample supernatant were mixed with 0.2 ml distilled water and 1.8 ml FRAP reagent and the absorbance of reaction mixture at 593 nm was measured spectrophotometrically after incubation at 37°C for 10 min. The 1 mmol/L FeSO_4 was used as the standard solution. The final result was expressed as the concentration of antioxidants having a ferric reducing ability equivalent to that of 1 mmol/L FeSO_4 . Adequate dilution was needed if the FRAP value measured was over the linear range of standard curve.

3. Results

3.1. FRAP values of fruit peel, pulp and seed fractions

The FRAP value of peel, pulp and seed fractions of 28 fruits is summarized in Table 1 on the basis of 100g wet weight. Among all fruit pulps tested, the hawthorn had the highest FRAP value, followed by date, guava, kiwifruit, purple mulberry, strawberry, white pomegranate, “Lukan” tangerine, “Honey” tangerine, orange, lemon, cherry, longan, “Banana flavored” apple, pineapple, banana, plum, lychee, kumquat, “Red rose” grape, pomelo, mango, “Jiubao” peach, apricot, “Hami” melon, “Duck” pear, “Jingxin #1” watermelon and persimmon. There was more than 90-fold difference between FRAP values in various fruit pulps.

Of the fruit peels analyzed, the pomegranate displayed the highest FRAP value, followed by hawthorn, date, kiwifruit, “Red rose” grape, guava, mango, plum, “Lukan” tangerine, orange, “Honey” tangerine, longan, “Banana flavored” apple, banana, lychee, cherry, lemon, pineapple, pomelo, “Jiubao” peach, “Duck” pear, apricot, persimmon, “Hami” melon, “Jingxin #1” watermelon and kumquat. There was more than 328-fold difference between FRAP values in various fruit peels.

For the fruit seeds, the grape showed the highest FRAP value, followed by longan, lychee, mango, guava, “Duck” pear, date, persimmon, “Jiubao” peach, “Lukan” tangerine, “Jingxin #1” watermelon, lemon, “Banana flavored” apple, cherry, apricot, white pomegranate, kumquat, plum, hawthorn and “Hami” melon. There was more than 179-fold difference between FRAP values in various fruit seeds.

In the comparison between all fruit peel, pulp and seed fractions tested, there were five fractions with FRAP value over 20 mmol/L. The pomegranate peels had the highest FRAP value, followed by grape seeds, hawthorn peels, longan seeds and lychee seeds. Except for kumquat, all fruit peels possessed higher FRAP values than their pulps. The FRAP value of most fruit seed fractions also was higher than that of fruit pulps with the exception of

Table 1

FRAP values of peel, pulp and seed fractions of 28 fruits (mmol/100g wet weight)

Fruits	Pulp	Peel	Seed	Total
Hawthorn	13.42 ± 0.74	29.25 ± 1.59	0.43 ± 0.03	43.10
Date	6.98 ± 0.29	16.69 ± 0.55	1.77 ± 0.13	25.44
Guava	6.07 ± 0.69	10.24 ± 0.24	4.71 ± 0.24	21.02
Kiwifruit	4.38 ± 0.20	11.13 ± 0.23	–	15.51
Purple mulberry	4.11 ± 0.25	–	–	4.11
Strawberry	3.29 ± 0.30	–	–	3.29
White pomegranate	3.10 ± 0.12	82.11 ± 4.01	0.72 ± 0.05	85.93
Lukan tangerine	2.29 ± 0.13	6.94 ± 0.40	1.15 ± 0.02	10.38
Honey tangerine	2.19 ± 0.08	5.44 ± 0.07	–	7.63
Orange	1.89 ± 0.19	5.69 ± 0.26	–	7.58
Lemon	1.43 ± 0.07	2.30 ± 0.12	0.91 ± 0.07	4.64
Cherry	0.99 ± 0.21	2.82 ± 0.29	0.77 ± 0.12	4.58
Longan	0.94 ± 0.05	3.98 ± 0.30	24.26 ± 2.79	29.18
Banana flavored apple	0.80 ± 0.05	3.24 ± 0.39	0.84 ± 0.09	4.88
Pineapple	0.80 ± 0.08	2.01 ± 0.03	–	2.81
Banana	0.73 ± 0.11	3.16 ± 0.16	–	3.89
Plum	0.71 ± 0.01	8.09 ± 0.55	0.65 ± 0.06	9.45
Lychee	0.59 ± 0.11	2.86 ± 0.12	22.36 ± 0.97	25.81
Kumquat	0.50 ± 0.05	0.25 ± 0.05	0.66 ± 0.03	1.41
Red rose grape	0.49 ± 0.04	11.02 ± 1.83	55.54 ± 1.62	67.05
Pomelo	0.39 ± 0.03	1.84 ± 0.04	–	2.23
Mango	0.38 ± 0.08	10.13 ± 0.37	14.59 ± 0.55	25.10
Jiubao peach	0.38 ± 0.03	0.95 ± 0.08	1.17 ± 0.03	2.50
Apricot	0.34 ± 0.07	0.79 ± 0.07	0.72 ± 0.09	1.85
Hami melon	0.24 ± 0.06	0.52 ± 0.07	0.31 ± 0.13	1.07
Duck pear	0.22 ± 0.03	0.89 ± 0.08	2.06 ± 0.09	3.17
Jingxin #1	0.16 ± 0.01	0.42 ± 0.11	1.01 ± 0.07	1.59
watermelon				
Persimmon	0.14 ± 0.03	0.62 ± 0.03	1.48 ± 0.08	2.24
Total	52.75	213.14	131.40	397.29

Data are expressed as mean ± SD. Each fruit was analyzed five times.

pomegranate, “Lukan” tangerine, lemon, cherry and plum. There was more than 586-fold difference between FRAP values in various fruit peel, pulp and seed fractions.

3.2. Percentage of fruit pulp FRAP value contributed by vitamin C

Data on vitamin C content of fruit pulps were cited from the China Food Composition 2002, which was compiled by Institute of Nutrition and Food Safety of China CDC [15]. As indicated in Table 2, the calculated contribution of vitamin C to the fruit pulp FRAP value varied considerably from fruit to fruit, ranging from 107.75% to 2.43%. In lychee, kumquat and persimmon pulps, over 80% FRAP value was from vitamin C contribution. In hawthorn, apple, pomegranate, “Lukan” tangerine, cherry, longan, banana, grape and apricot pulps, vitamin C contribution was less than 20%. There were 19 fruits in this study, in which more than 50% FRAP value of pulp fraction was not contributed by vitamin C as calculated.

Table 2
Percentage of fruit pulp FRAP value contributed by vitamin C

Fruits	VC content ^a (mg%)	Calculated FRAP value ^b (mmol)	Percentage of fruit pulp FRAP value (%)
Hawthorn	53	0.80	5.96
Date	243	3.67	52.58
Guava	68	1.03	16.97
Kiwifruit	62	0.94	21.46
Strawberry	47	0.71	21.58
White pomegranate	5	0.08	2.58
Lukan tangerine	19	0.29	12.66
Honey tangerine	19	0.29	13.24
Orange	33	0.50	26.46
Lemon	22	0.33	23.08
Cherry	10	0.15	15.15
Longan	12	0.18	19.15
Banana flavored apple	3	0.05	6.25
Pineapple	18	0.27	33.75
Banana	8	0.12	16.44
Plum	16	0.24	33.80
Lychee	41	0.62	105.08
Kumquat	35	0.53	106.0
Red rose grape	4	0.06	12.24
Pomelo	23	0.35	89.74
Mango	23	0.35	92.11
Jiubao peach	8	0.12	31.58
Apricot	4	0.06	17.65
Hami melon	12	0.18	75.0
Duck pear	4	0.06	27.27
Jingxin #1 watermelon	7	0.11	68.75
Persimmon	10	0.15	107.14

^a Cited from China Food Composition 2002, compiled by Institute of Nutrition and Food Safety, China CDC. Beijing: Peking University Medical Press, 2002, p 82-97. The vitamin C content of purple mulberry is not available.

^b 100mg Vitamin C is equivalent to 1.51mmol FRAP value as determined.

4. Discussion

There are many different antioxidants contained in fruits and it is very difficult to measure each antioxidant component separately. Therefore, several methods have been developed to evaluate the total antioxidant activity of fruits or other plants and animal tissues. Among them, Trolox equivalent antioxidant capacity [16], total radical absorption potentials [17], oxygen radical absorption capacity assays [18,19] are the representative methods frequently used in various investigations. However, none of the methods mentioned above can be treated as a total antioxidant capacity assay because what they really measure is the capacity of antioxidants in scavenging specific radicals, inhibiting lipid peroxidation or chelating metal ions. Another method commonly used is the FRAP assay [14]. It can be used to measure the total reducing capability of antioxidants based on the reaction principle described in the materials and methods section. We selected the FRAP assay to evaluate the antioxidant activities of peel, pulp and seed fractions of fruits for following reasons. First, the

FRAP assay treats the antioxidants in the samples as reductants in a redox-linked colorimetric reaction. Second, the procedure of FRAP assay is relatively simple and easy to be standardized. One possible disadvantage with FRAP assay is the fact that this assay does not react fast with some antioxidants, such as glutathione. However, we consider that FRAP assay is still suitable for assessment of antioxidant activity of fruit samples because only limited amounts of plant glutathione are absorbed by humans [20].

The antioxidant activity of fruit pulps had been determined previously by others using different methods [21,22,23,24]. Halvorsen et al also assessed the total antioxidant concentrations of some fruits by FRAP assay [25]. We tried to correlate the fruit FRAP values obtained in this study with the data reported by others and found no positive relationship between them. It is not surprising that different methods would give different results because they are based on different principles, as discussed above. Although Halvorsen et al used the same FRAP assay principle, discrepancy may still exist because the antioxidant activity of fruits could be influenced by the geographical origin, cultivar and harvest or storage time [26]. Actually, the data by Halvorsen et al also indicated clearly that fruit samples collected from different geographic regions in the world varied in antioxidant capacity as measured by FRAP assay. For example, the antioxidant capacity of orange samples was ranged from 0.83 mmol/L to 1.50 mmol/L as they were collected from Spain, Holland and Zenta respectively. The FRAP value of cherry samples was ranged from 0.62 mmol/L (USA sample) to 1.42 mmol/L (Norway sample) [25].

It had been documented previously that antioxidant activity of fruit pulps did not correlate with the vitamin C content [21,22]. Our data showed that contribution of vitamin C to the FRAP value of fruit pulps varied greatly among different fruits. We did not measure the vitamin C content of fruit pulps directly, but cited the data from China Food Composition 2002, an authoritative database for food composition in China. Therefore, it should be pointed out that though this database represents a diverse sampling and would not expect several fold fluctuation, direct analysis of vitamin C content will generate more precise estimation. It was impressive based on our calculation in current study that there were 19 fruits, in which more than 50% pulp FRAP value was not from vitamin C contribution. Thus, the antioxidants other than vitamin C may play a more important role in the antioxidant activity of these fruit pulps. The fruit with high antioxidant activity generally contains more antioxidants and most of these antioxidants had been showed to be phenolic acids and flavonoids in nature [11].

The majority of fruit peels exhibited 2 to 27-fold higher antioxidant activity than the fruit pulps as measured in the present study. It cannot be explained totally by the difference between peel and pulp fractions in water content. We compared the water content of peel and pulp fractions of apple, orange, grape, banana and pomegranate (data not shown) and found that peel fractions contained only about 10% less moisture than pulp fractions. Therefore, the fruit peel fractions, particularly those with much high antioxidant activity may be rich sources of antioxidants and deserve for further study. Singh et al had reported that the extracts of pomegranate peel exhibited higher antioxidant activity *in vitro* as compared to the seed extracts^[13]. Further report from the same laboratory showed that pomegranate peel extracts could protect significantly against CCl₄ toxicity on the liver, in which the ROS damages were involved [27].

Some fruit seed fractions also displayed higher antioxidant activity than fruit pulp fractions. The grape seeds had the highest FRAP value among all fruit seeds tested. The grape seeds had been proved to be rich in proanthocyanidins, which could protect against the development of atherosclerosis in cholesterol-fed rabbits [12,28]. Besides the grape seeds, longan and lychee seeds also presented relatively high antioxidant activity in the present study. However, no data could be found so far in the study of longan and lychee seeds for their antioxidant activities in vitro or in vivo in the literature.

In summary, the antioxidant activities of peel, pulp and seed fractions of 28 fruits have been compared using the FRAP assay in this study. To our knowledge, the study reported here is the most comprehensive comparison on the antioxidant activity among different fruit fractions. Some fruit peel and seed fractions have strong antioxidant activity and may be rich sources of antioxidants. Further studies on the effective antioxidants contained in these fruit fractions and the mechanisms by which they protect against disease development are highly warranted.

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